Arduino UNO

```
int pressed=digitalRead(button);

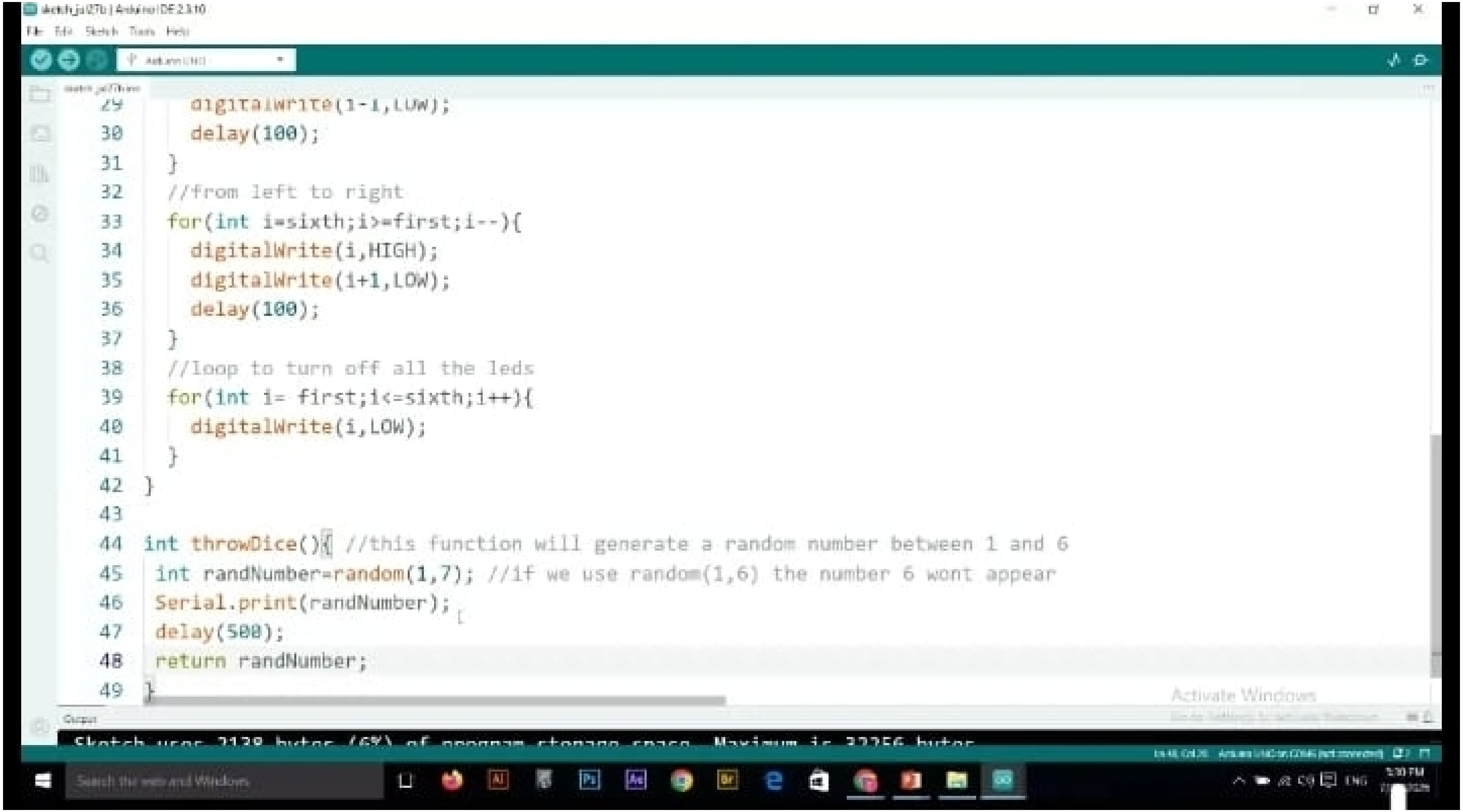
if(pressed==HIGH){
  buildUpTension();
  delay(100);
  int number=throwDice();
  showNumber(number);
}
}

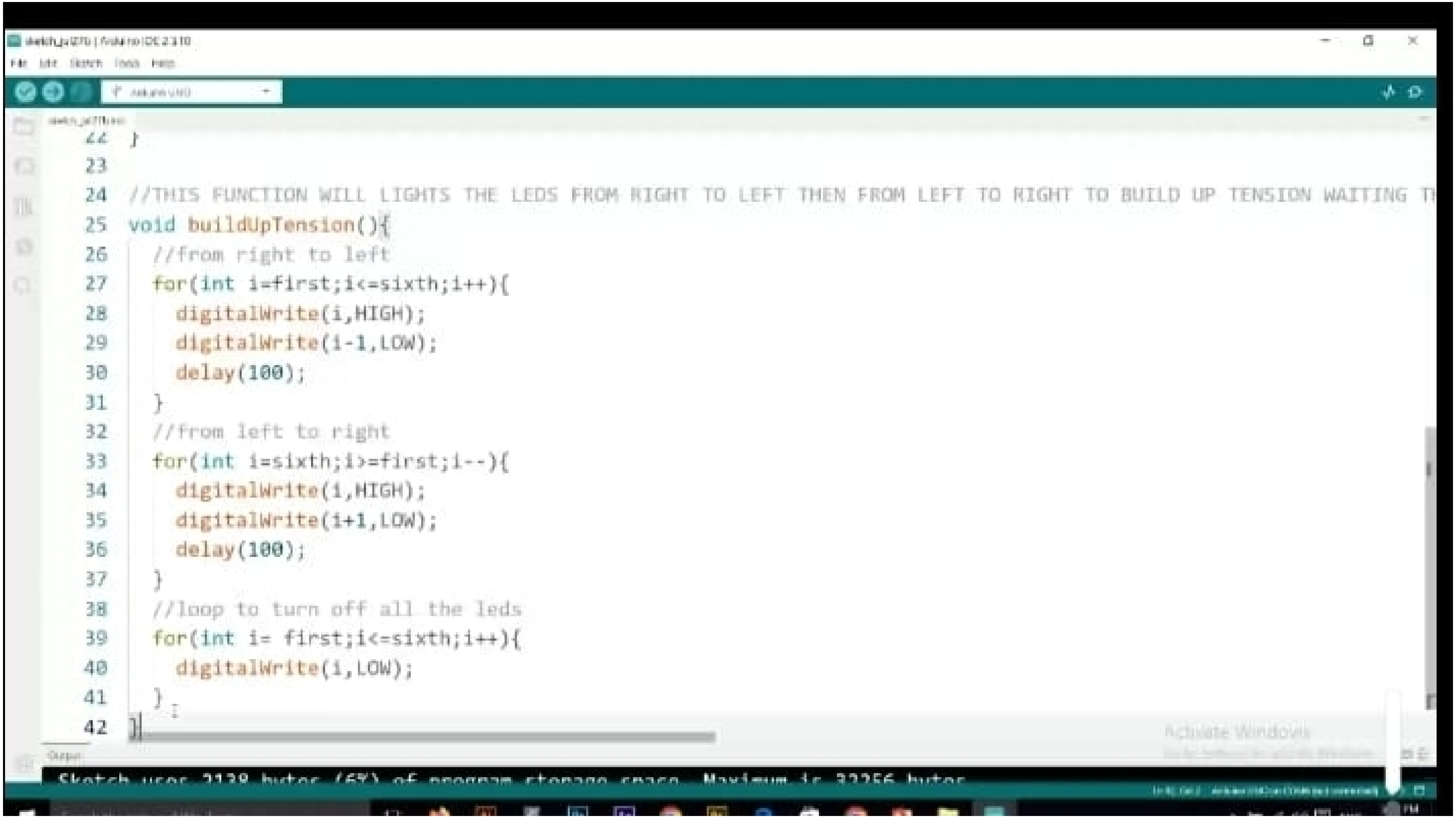
//THIS FUNCTION WILL LIGHTS THE LEDS FROM RIGHT TO LEFT THEN FROM LEFT TO
void buildUpTension(){
  //from right to left
  for(int i=first;i<=sixth;i++){
    digitalWrite(i,HIGH);
    digitalWrite(i-1,LOW);
    delay(100);
  }
  //from left to right
  for(int i=sixth;i>=first;i--){
    digitalWrite(i,HIGH);
```

Output

```
Sketch uses 2138 bytes (6%) of program storage space. Maximum is 32256 bytes.
```

Search the web and Windows


Arduino UNO
digitalWrite(i-1,LOW);
delay(100);
}
//from left to right
for(int i=sixth;i>=first;i--){
digitalWrite(i,HIGH);
digitalWrite(i+1,LOW);
delay(100);
}
//loop to turn off all the leds
for(int i= first;i<=sixth;i++){
digitalWrite(i,LOW);
}
}
int throwDice(){ //this function will generate a random number between 1 and 6
int randNumber=random(1,7); //if we use random(1,6) the number 6 wont appear
Serial.print(randNumber);
delay(500);
return randNumber;
}
Activate Windows
Sketch uses 2138 bytes (6%) of program storage space. Maximum is 32256 bytes.
Search the web and Windows
ENG


}
//THIS FUNCTION WILL LIGHTS THE LEDS FROM RIGHT TO LEFT THEN FROM LEFT TO RIGHT TO BUILD UP TENSION WAITING T
void buildUpTension(){
//from right to left
for(int i=first;i<=sixth;i++){
digitalWrite(i,HIGH);
digitalWrite(i-1,LOW);
delay(100);
}
//from left to right
for(int i=sixth;i>=first;i--){
digitalWrite(i,HIGH);
digitalWrite(i+1,LOW);
delay(100);
}
//loop to turn off all the leds
for(int i= first;i<=sixth;i++){
digitalWrite(i,LOW);
}
}
Sketch uses 2138 bytes (6%) of program storage space. Maximum is 32256 bytes.
Activate Windows

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    digitalWrite(i,HIGH);
    digitalWrite(i+1,LOW);
    delay(100);
  }
  //loop to turn off all the leds
  for(int i= first;i<=sixth;i++){
    digitalWrite(i,LOW);
  }
}
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//THIS FUNCTION WILL LIGHTS THE LEDS FROM RIGHT TO LEFT THEN FROM LEFT TO RIGHT TO BUILD UP TENSION WAITING TH
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    delay(100);
  }
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    digitalWrite(i,HIGH);
    digitalWrite(i+1,LOW);
    delay(100);
  }
  //loop to turn off all the leds
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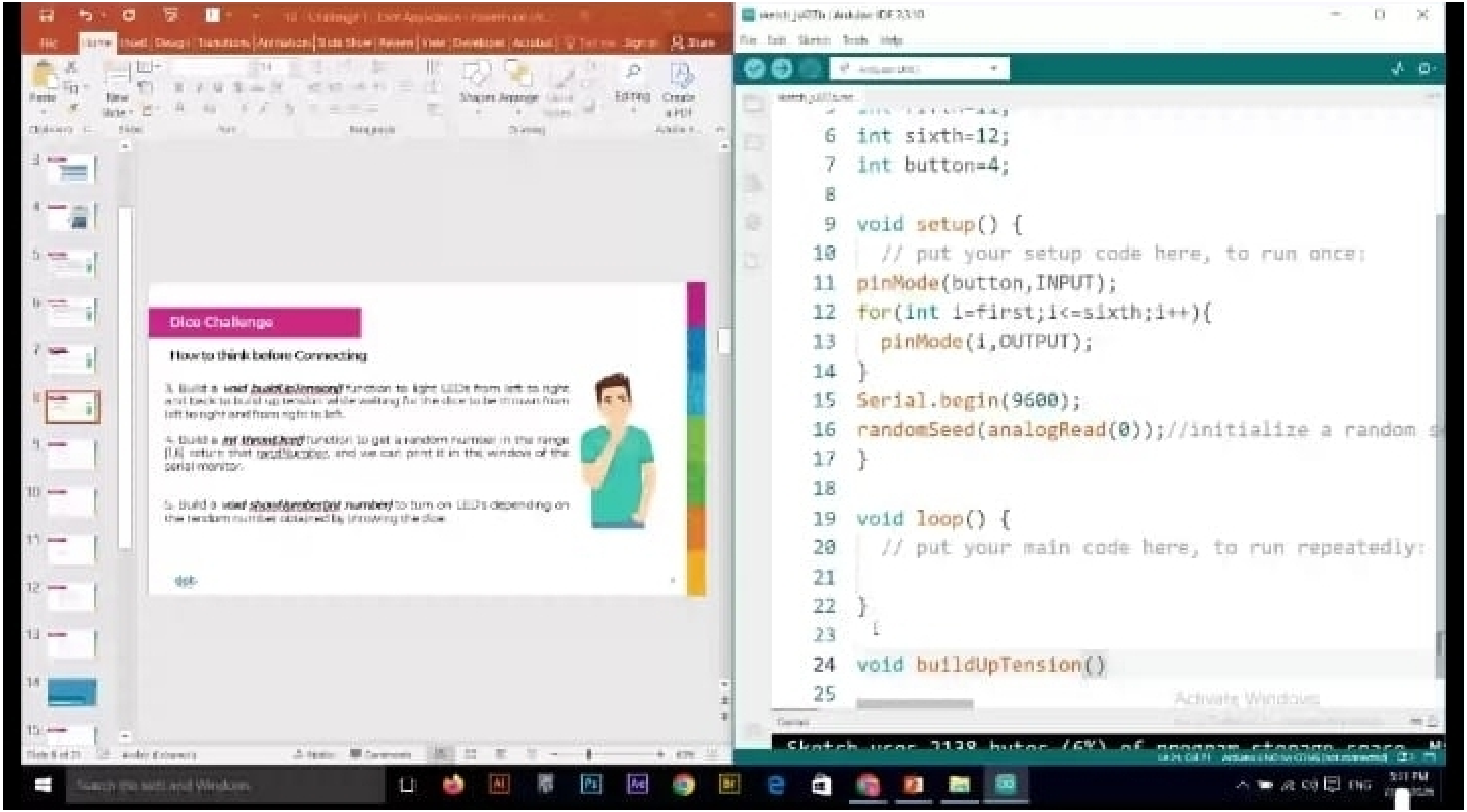
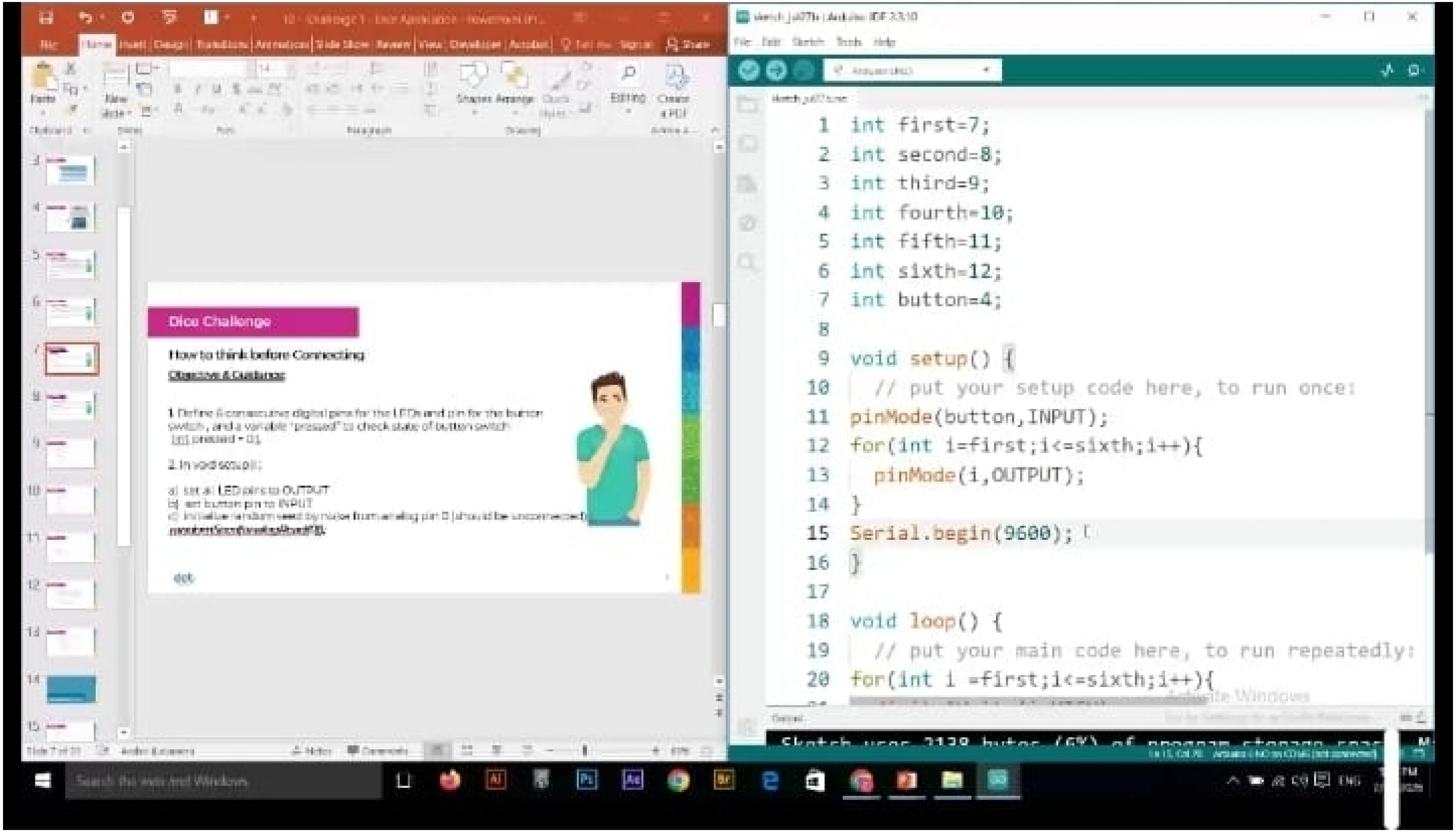
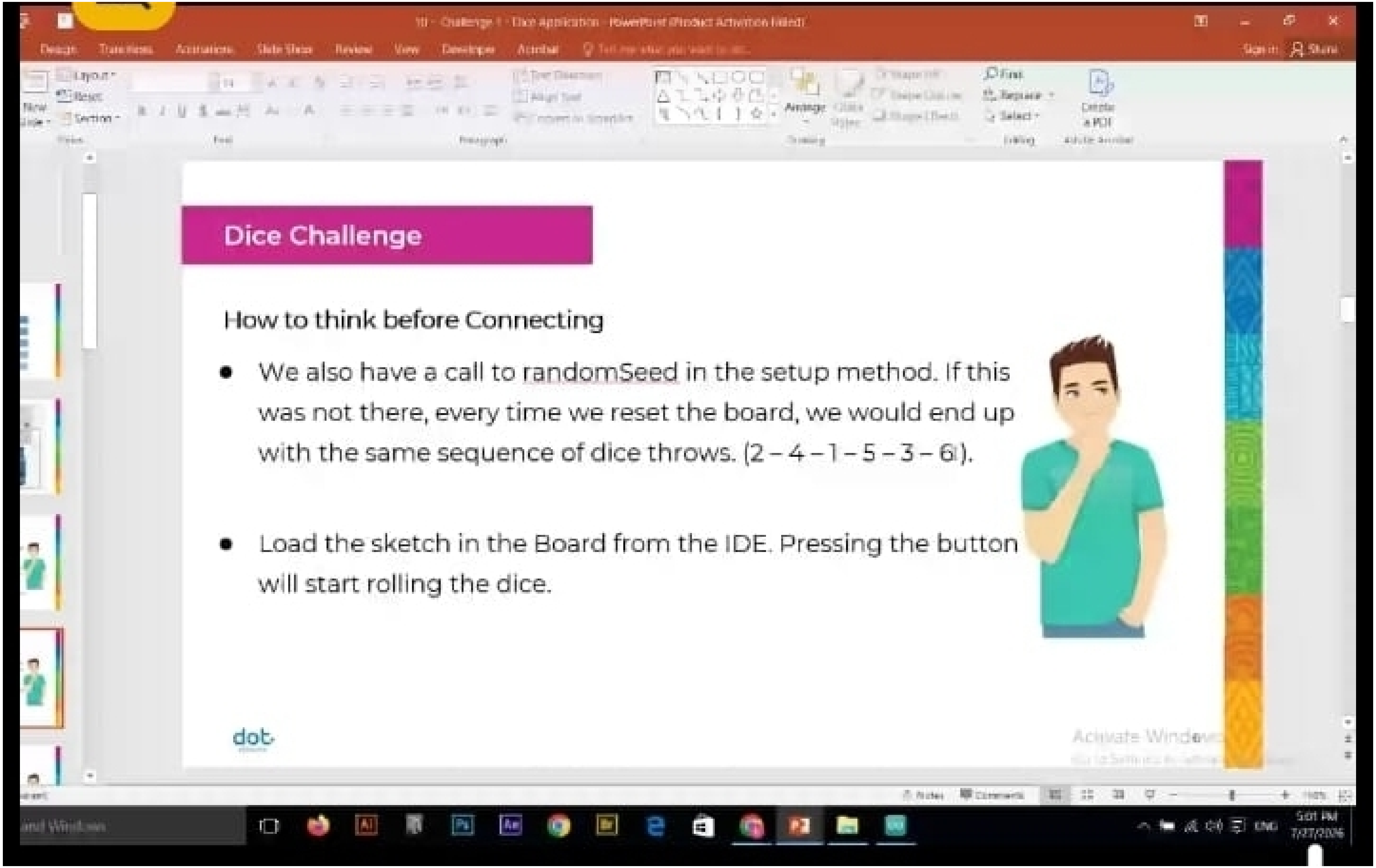
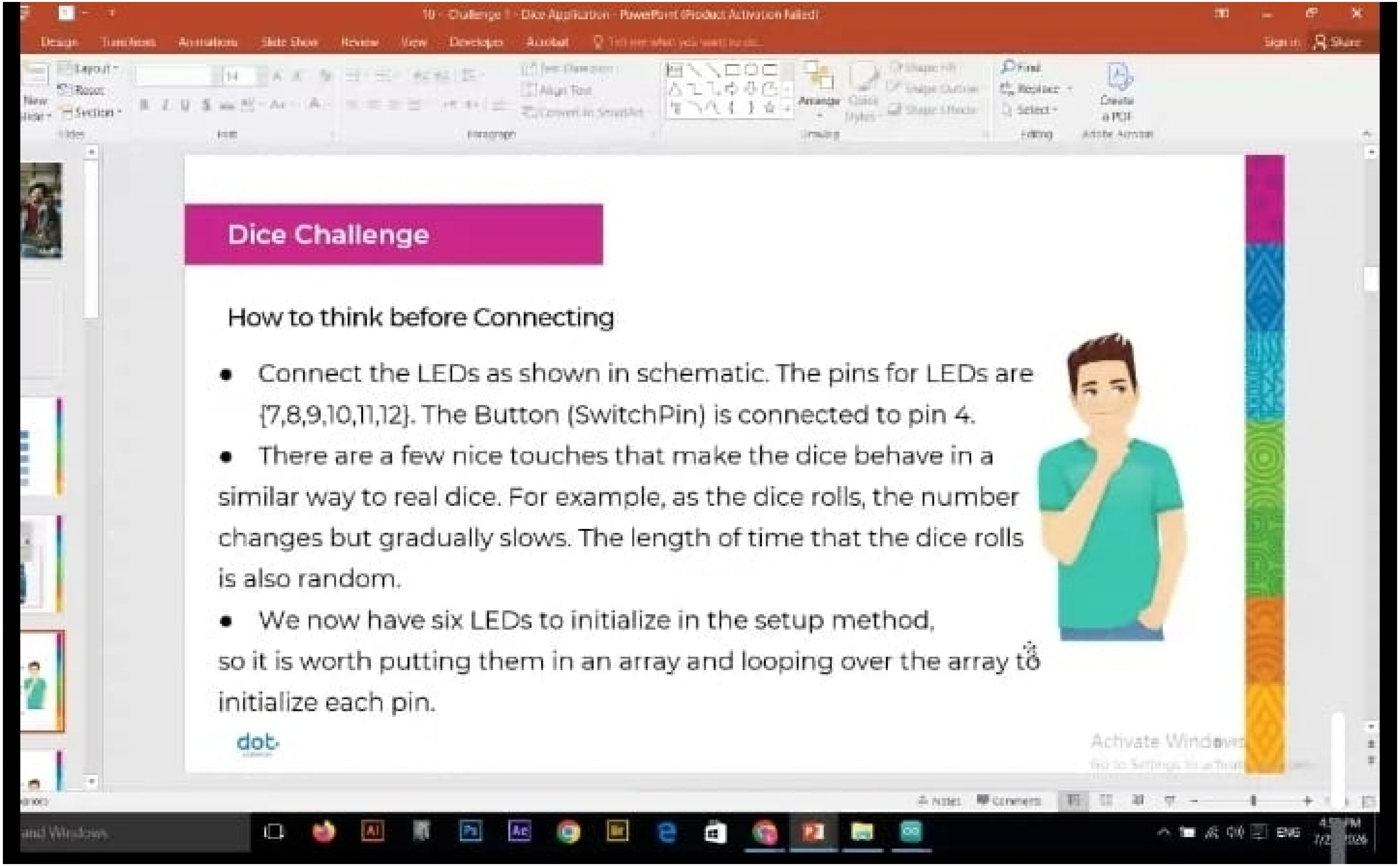
    digitalWrite(i,LOW);
  }
}
```

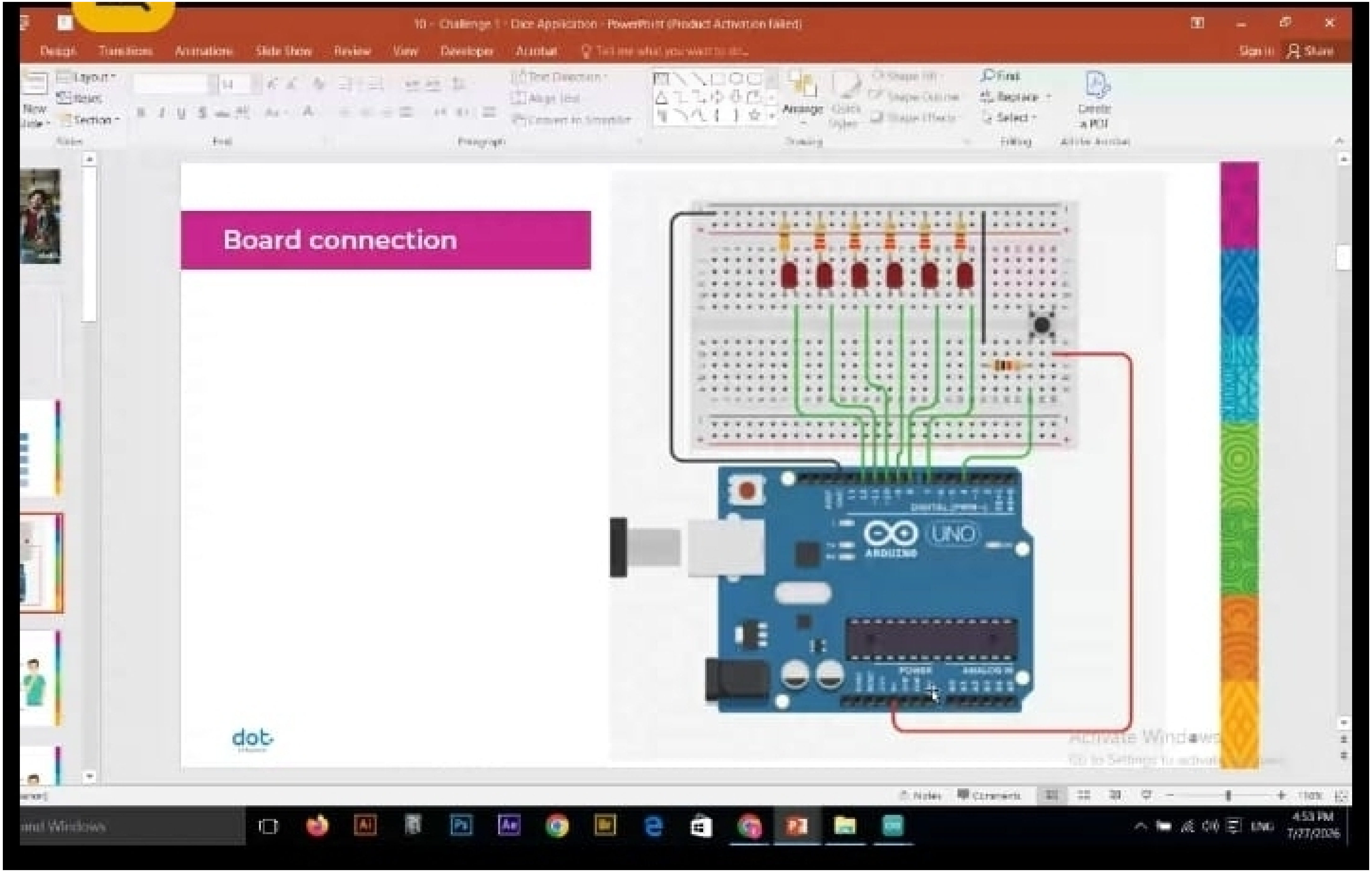
```
Sketch uses 2138 bytes (6%) of program storage space. Maximum is 32256 bytes.
```

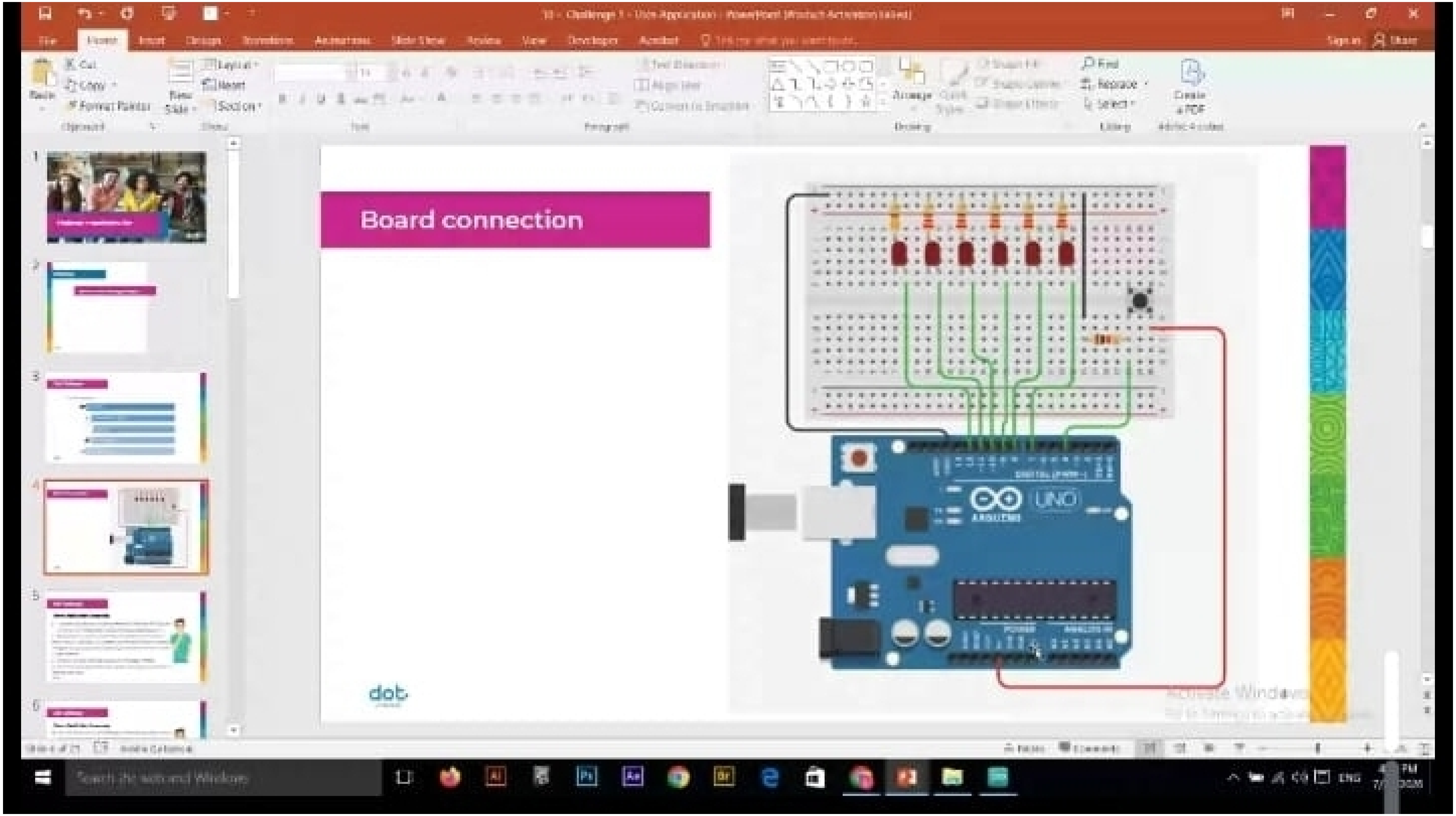

Dice Challenge
How to think before Connecting
int sixth=12;
int button=4;
void setup() {
// put your setup code here, to run once:
pinMode(button,INPUT);
for(int i=first;i<=sixth;i++){
pinMode(i,OUTPUT);
}
Serial.begin(9600);
randomSeed(analogRead(0));//initialize a random
}
void loop() {
// put your main code here, to run repeatedly:
}
void buildUpTension()
Activate Windows

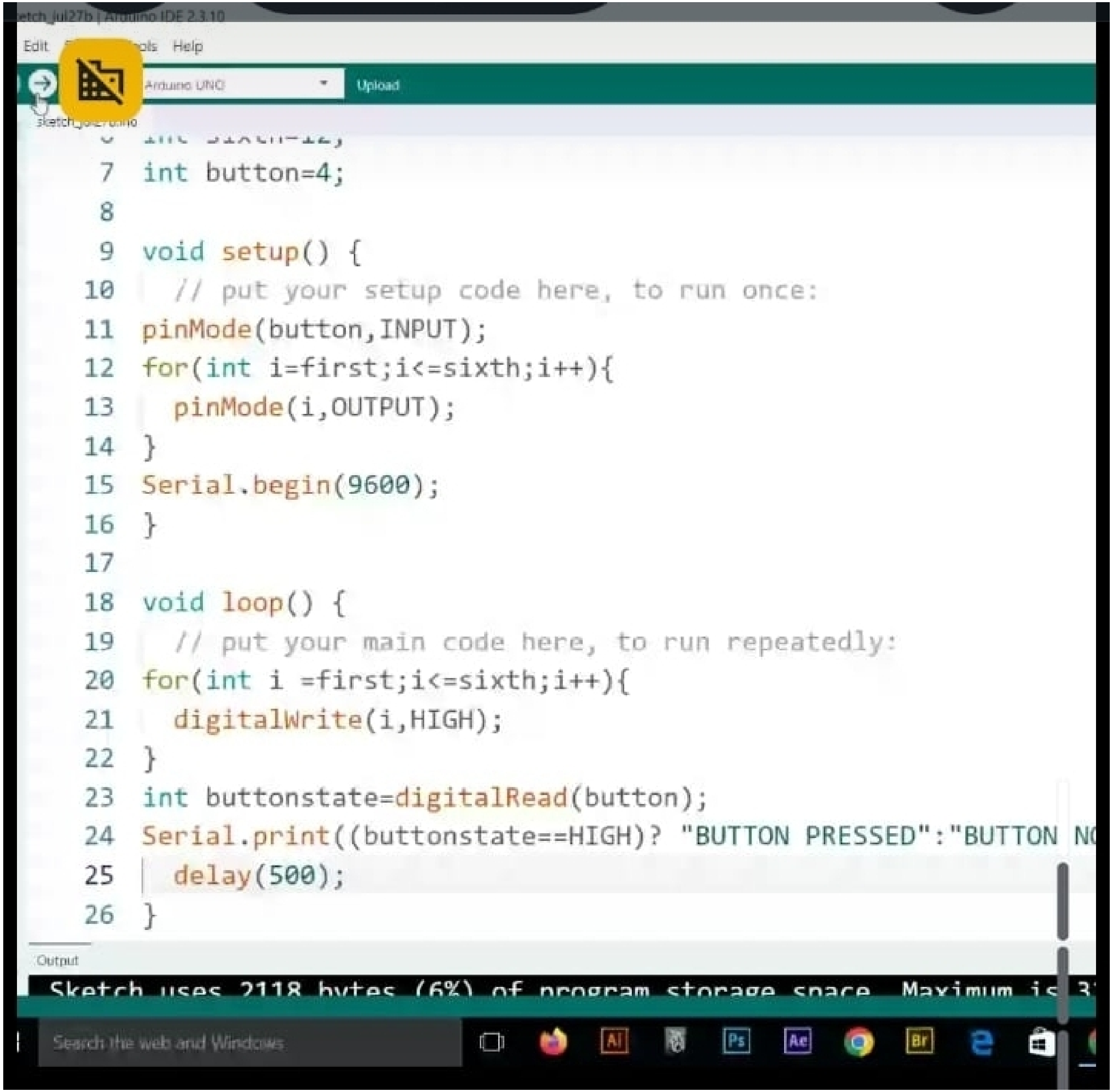

Dice Challenge
How to think before Connecting
a) set all LED pins to OUTPUT
b) set button pin to INPUT
int first=7;
int second=8;
int third=9;
int fourth=10;
int fifth=11;
int sixth=12;
int button=4;
void setup() {
// put your setup code here, to run once:
pinMode(button,INPUT);
for(int i=first;i<=sixth;i++){
pinMode(i,OUTPUT);
}
Serial.begin(9600);
}
void loop() {
// put your main code here, to run repeatedly:
for(int i =first;i<=sixth;i++){

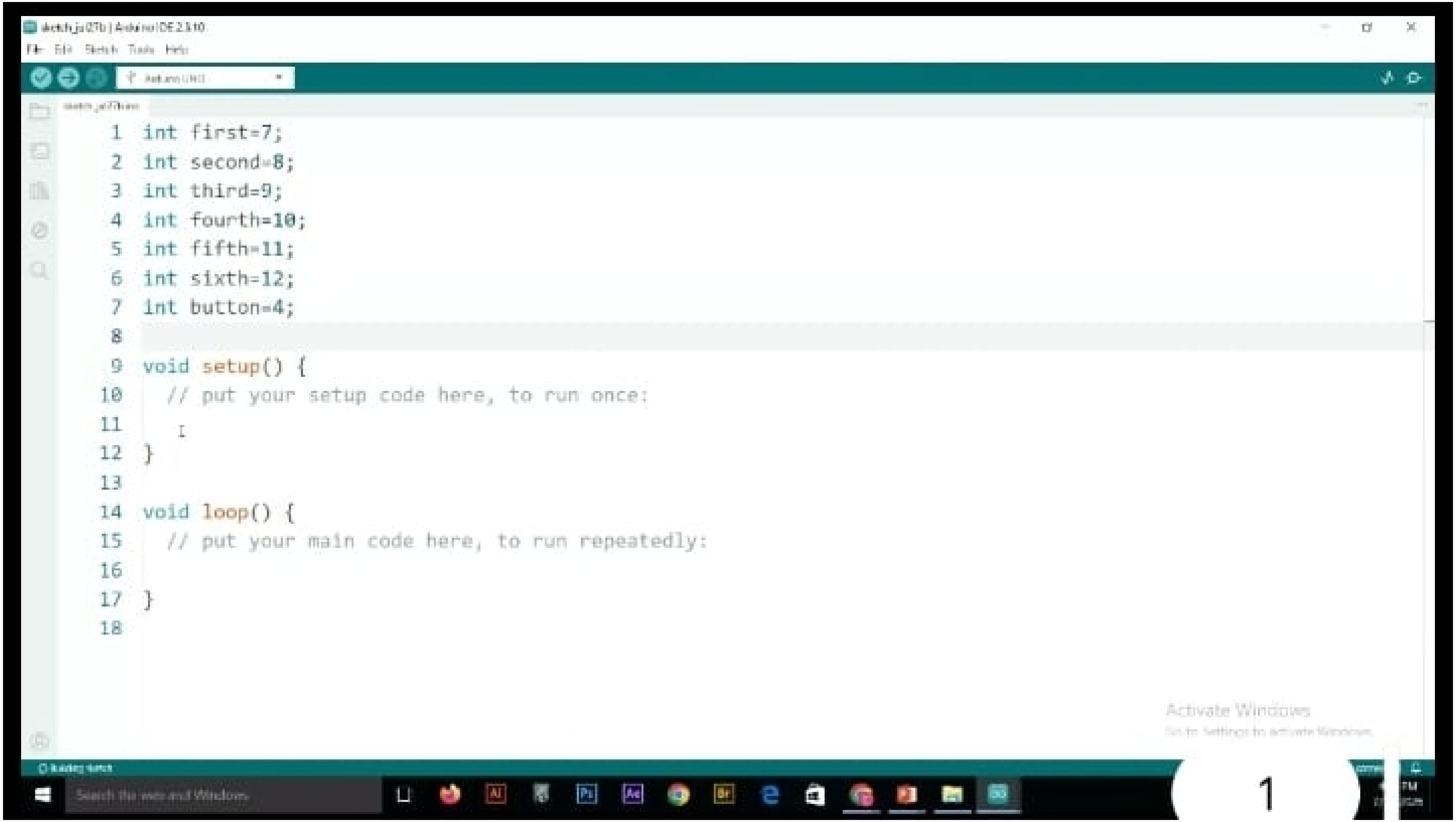

Dice Challenge
How to think before Connecting
We also have a call to randomSeed in the setup method. If this was not there, every time we reset the board, we would end up with the same sequence of dice throws. (2 – 4 – 1 – 5 – 3 – 6).
Load the sketch in the Board from the IDE. Pressing the button will start rolling the dice.
dot

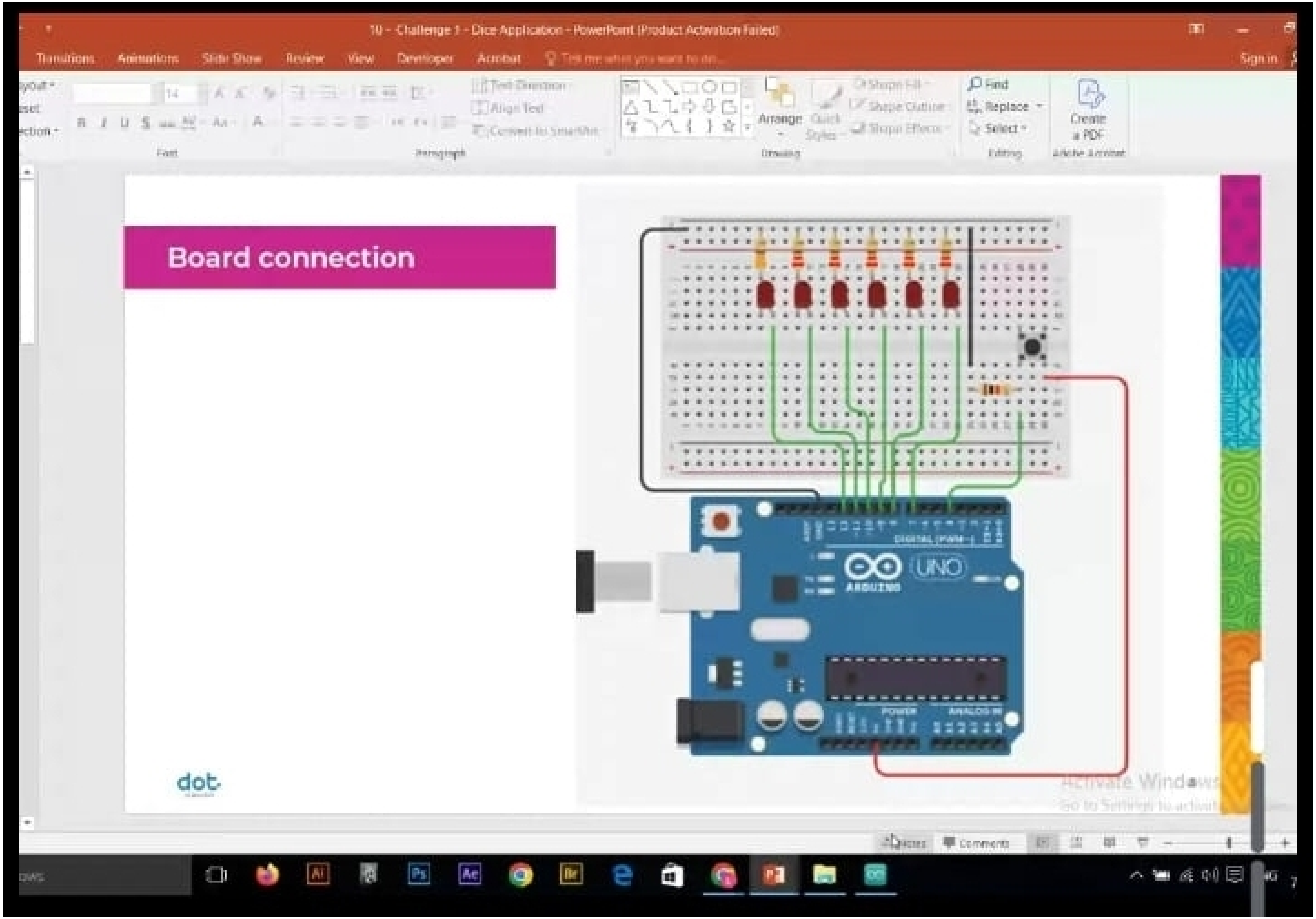

Dice Challenge
How to think before Connecting
Connect the LEDs as shown in schematic. The pins for LEDs are {7,8,9,10,11,12}. The Button (SwitchPin) is connected to pin 4.
There are a few nice touches that make the dice behave in a similar way to real dice. For example, as the dice rolls, the number changes but gradually slows. The length of time that the dice rolls is also random.
We now have six LEDs to initialize in the setup method, so it is worth putting them in an array and looping over the array to initialize each pin.
Activate Windows


10 - Challenge 1 - Dice Application - PowerPoint (Product Activation failed)
Design
Transitions
Animations
Slide Show
Review
View
Developer
Acrobat
Sign in
Share
Layout
Reset
Section
Find
Replace
Select
Arrange
Board connection
UNO
ARDUINO
dot.
Activate Windows
4:53 PM

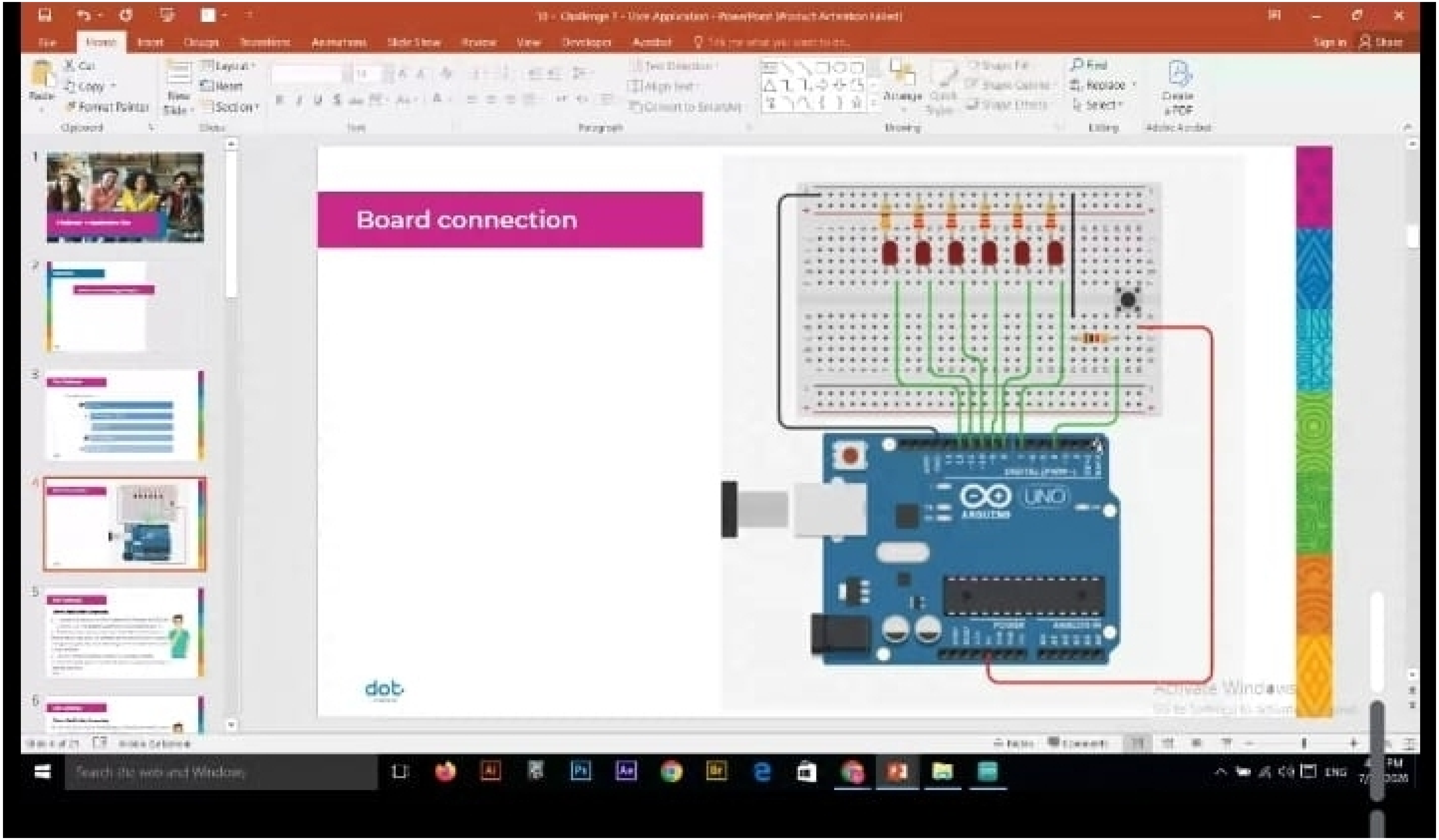

Board connection
dot
UNO
ARDUINO
Cut
Copy
Format Painter
Layout
Reset
Section
Arrange
Find
Replace
Select
Create a PDF

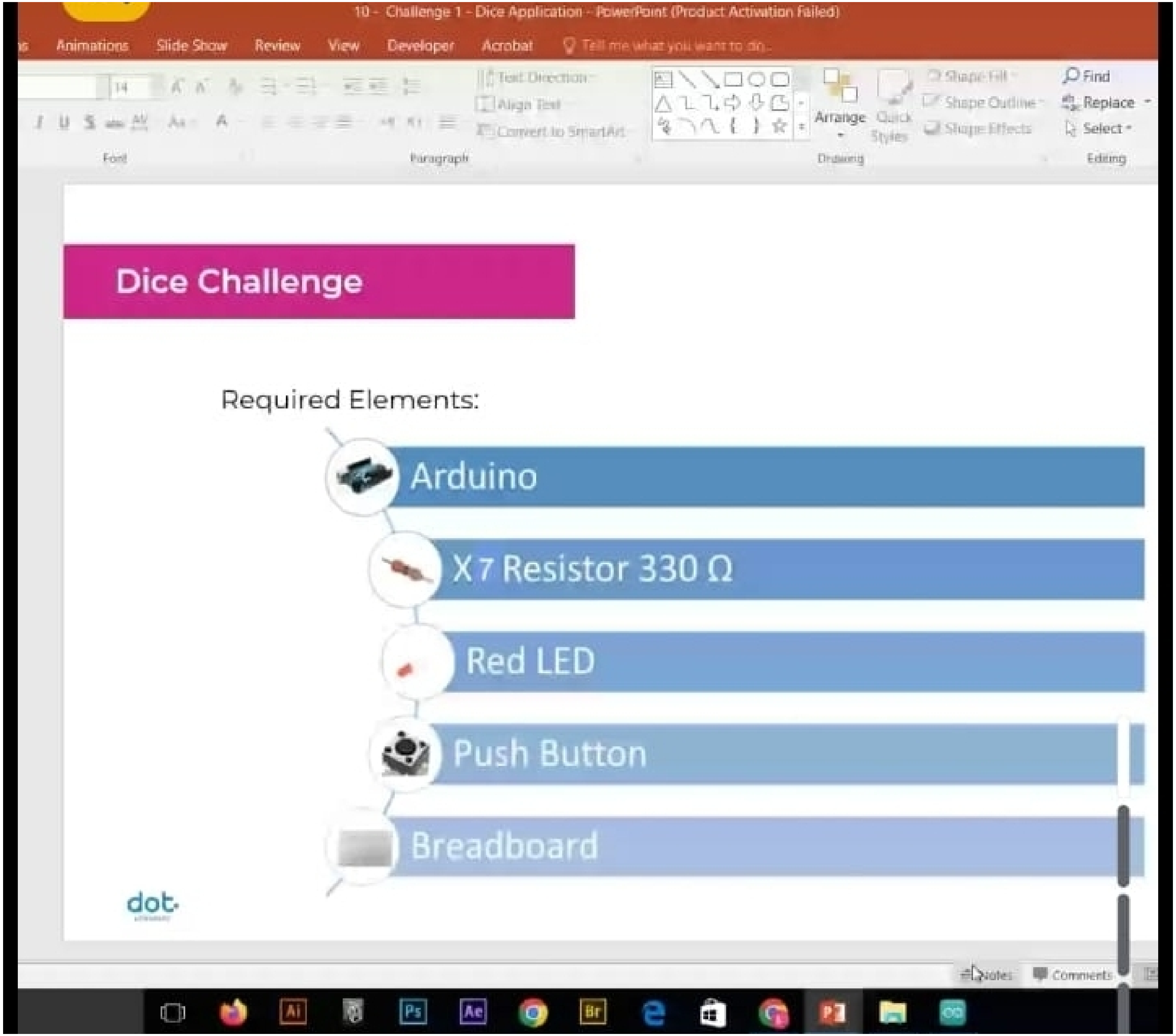

Edit
Help
Arduino UNO
Upload
int button=4;
void setup() {
// put your setup code here, to run once:
pinMode(button,INPUT);
for(int i=first;i<=sixth;i++){
pinMode(i,OUTPUT);
}
Serial.begin(9600);
}
void loop() {
// put your main code here, to run repeatedly:
for(int i =first;i<=sixth;i++){
digitalWrite(i,HIGH);
}
int buttonstate=digitalRead(button);
Serial.print((buttonstate==HIGH)? "BUTTON PRESSED":"BUTTON N
delay(500);
}
Output
Search the web and Windows

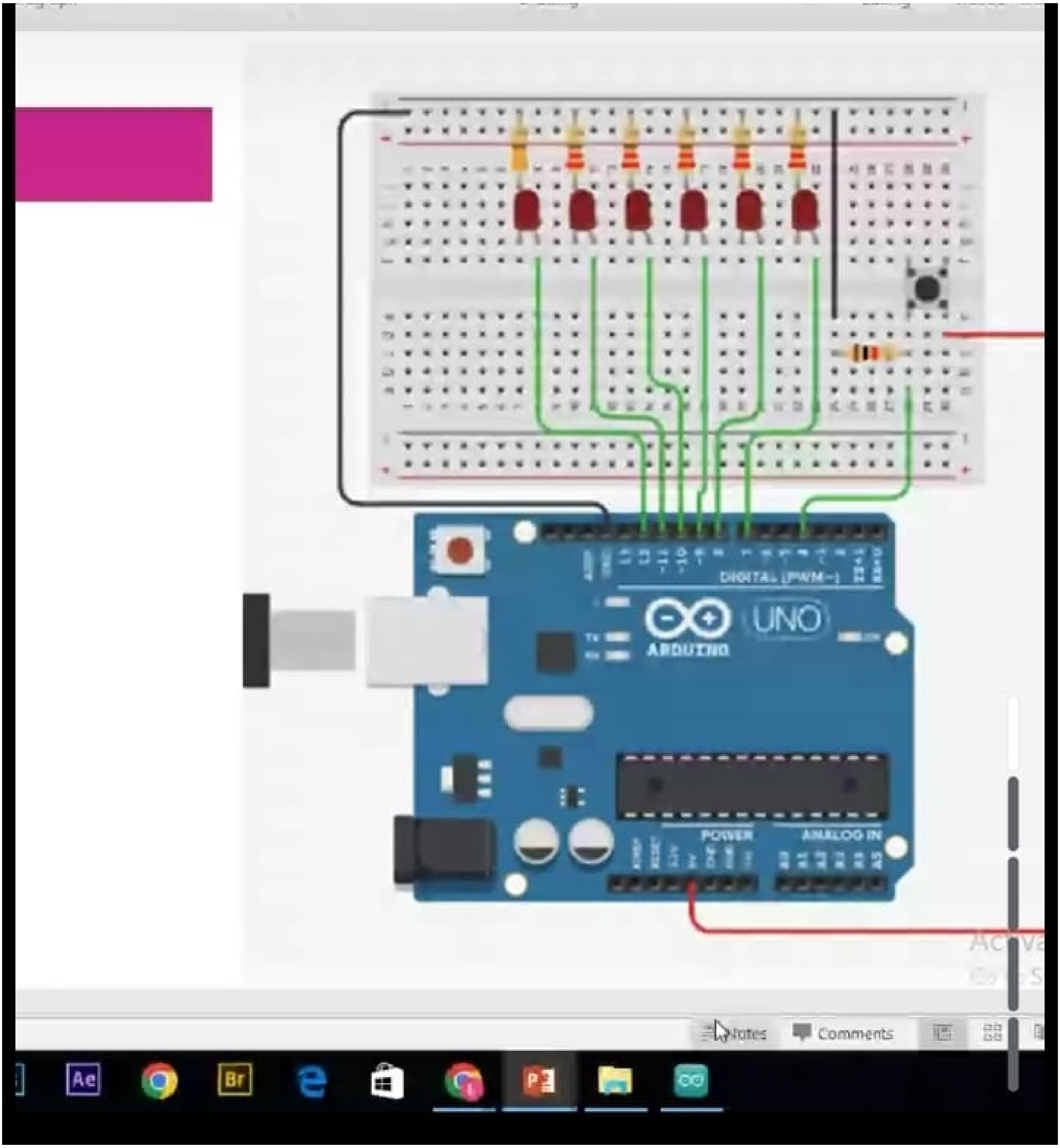

File Edit Sketch Tools Help
int first=7;
int second=8;
int third=9;
int fourth=10;
int fifth=11;
int sixth=12;
int button=4;
void setup() {
// put your setup code here, to run once:
}
void loop() {
// put your main code here, to run repeatedly:
}
Activate Windows
1

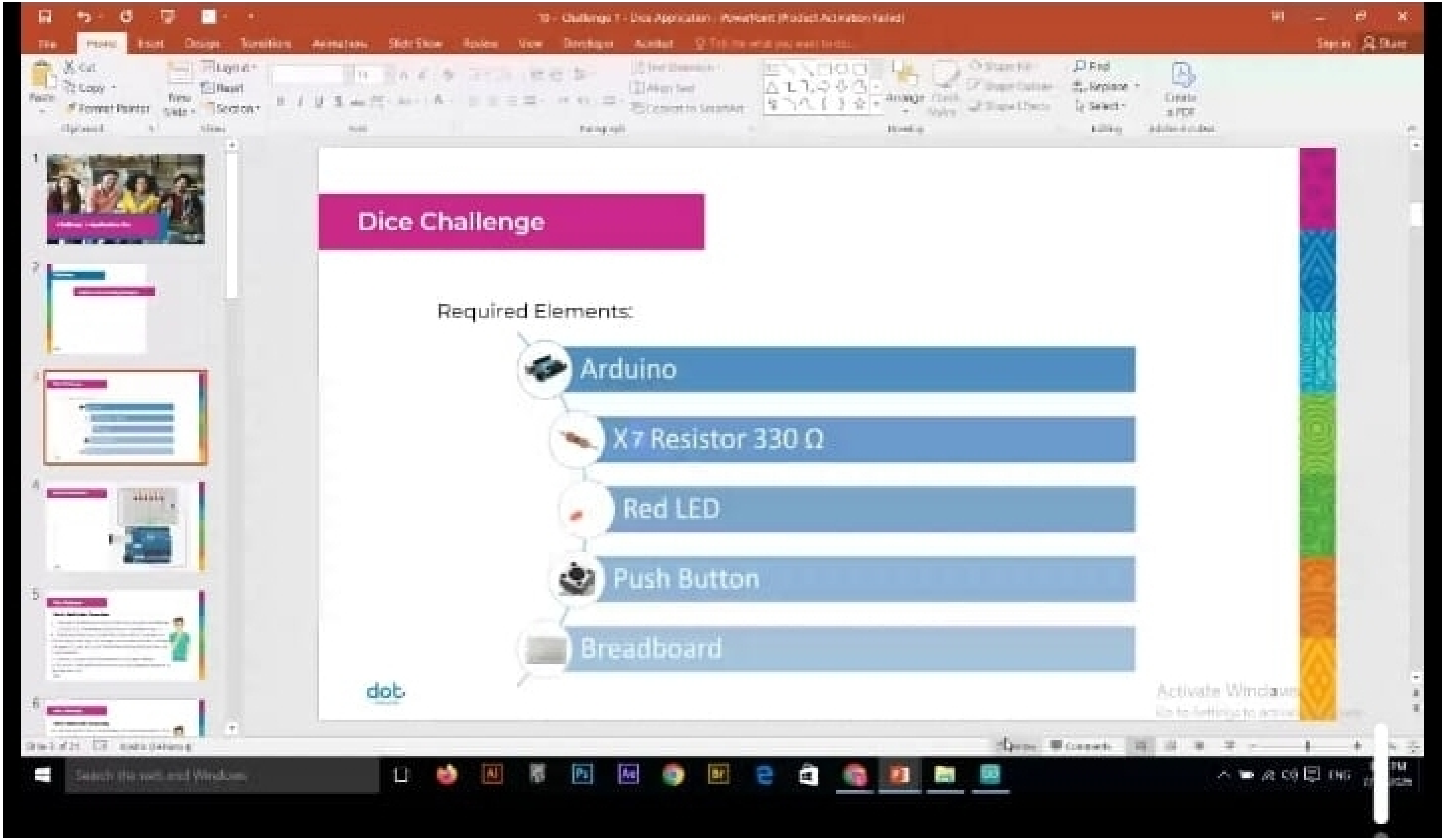


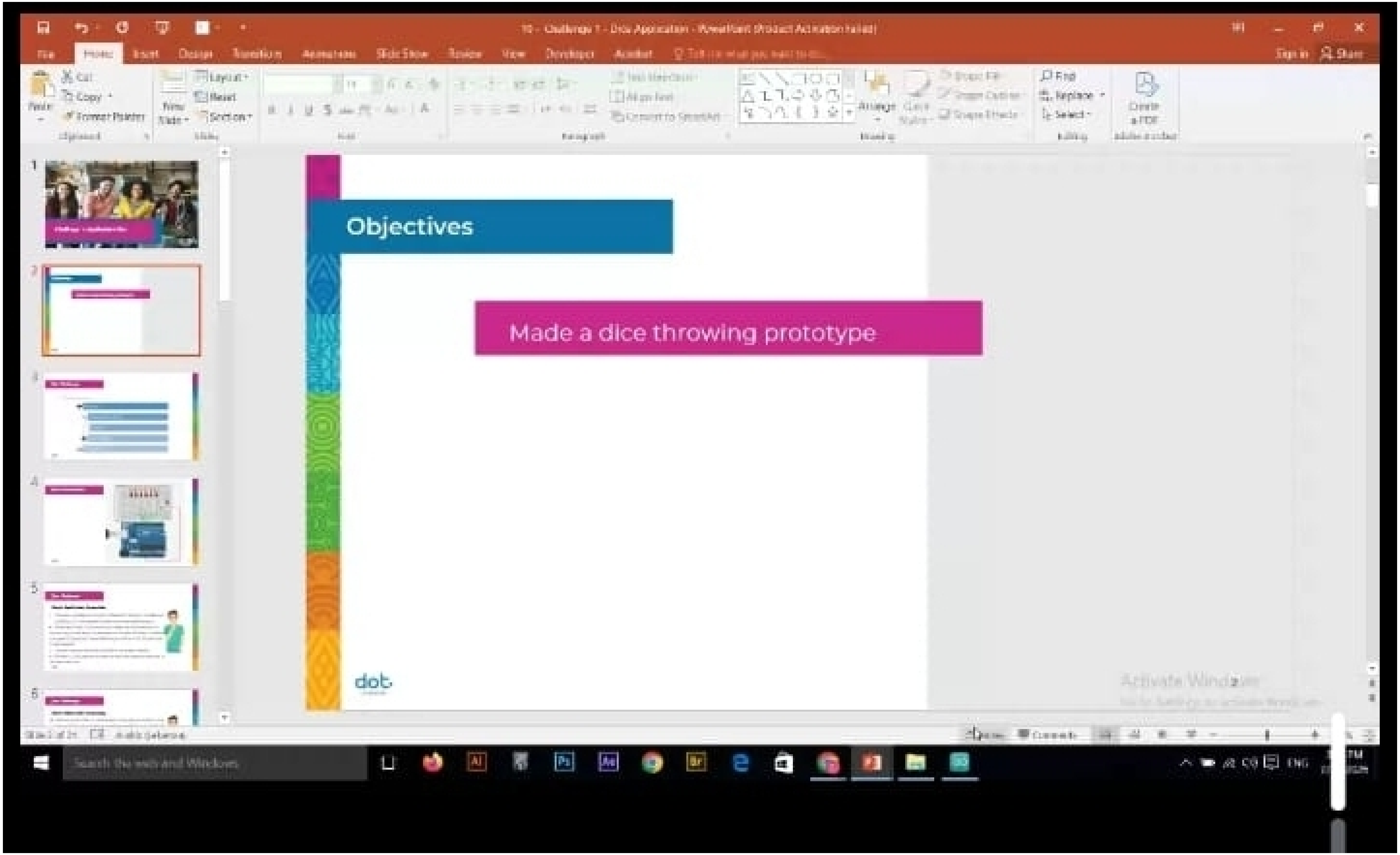
10 - Challenge 1 - Dice Application - PowerPoint (Product Activation Failed)
Transitions
Animations
Slide Show
Review
View
Developer
Acrobat
Sign in
Arrange
Find
Replace
Select
Create a PDF
Board connection
dot.
DIGITAL (PWM~)
UNO
ARDUINO
POWER
ANALOG IN
Activate Windows
Comments

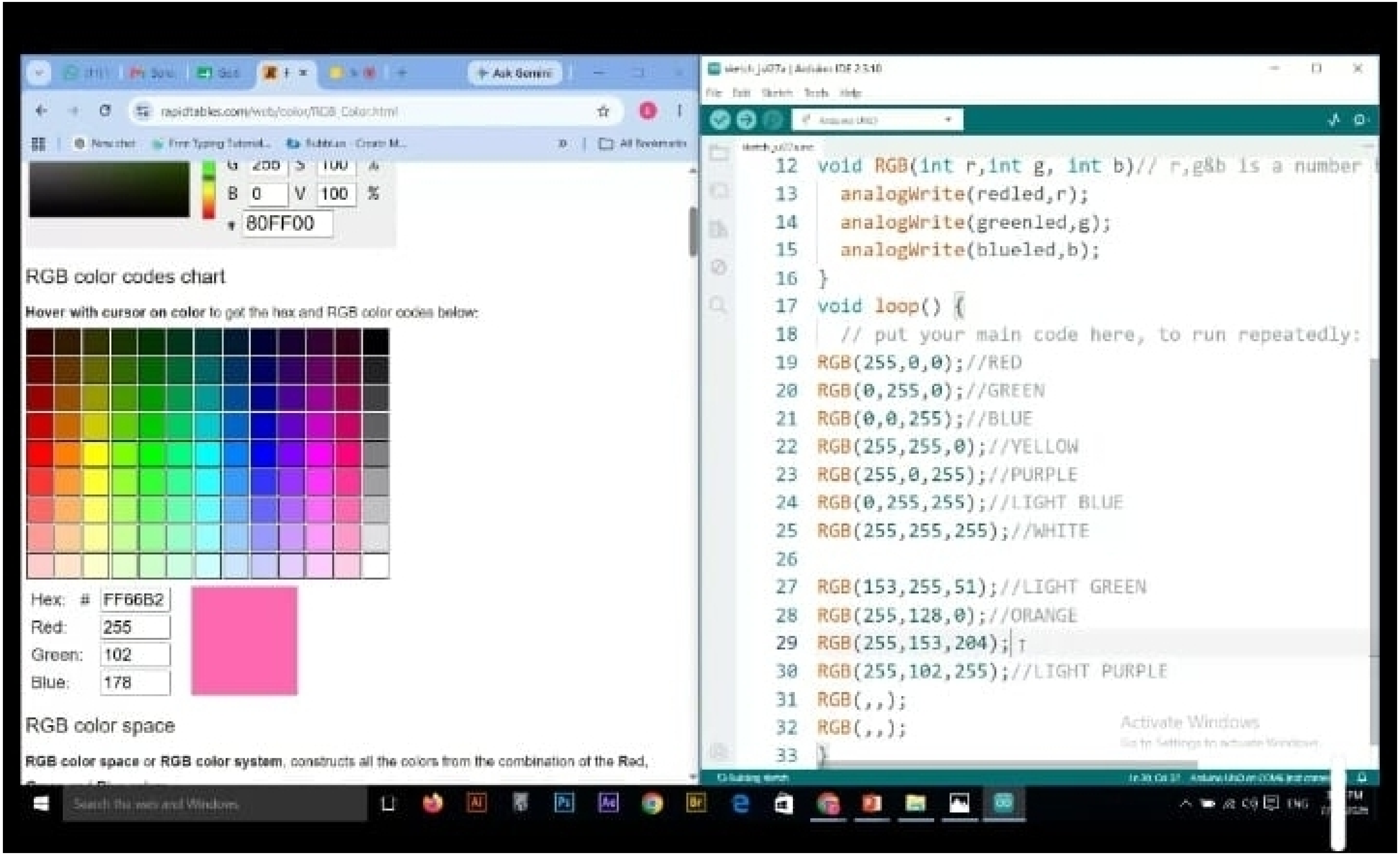

Board connection
dot
UNO

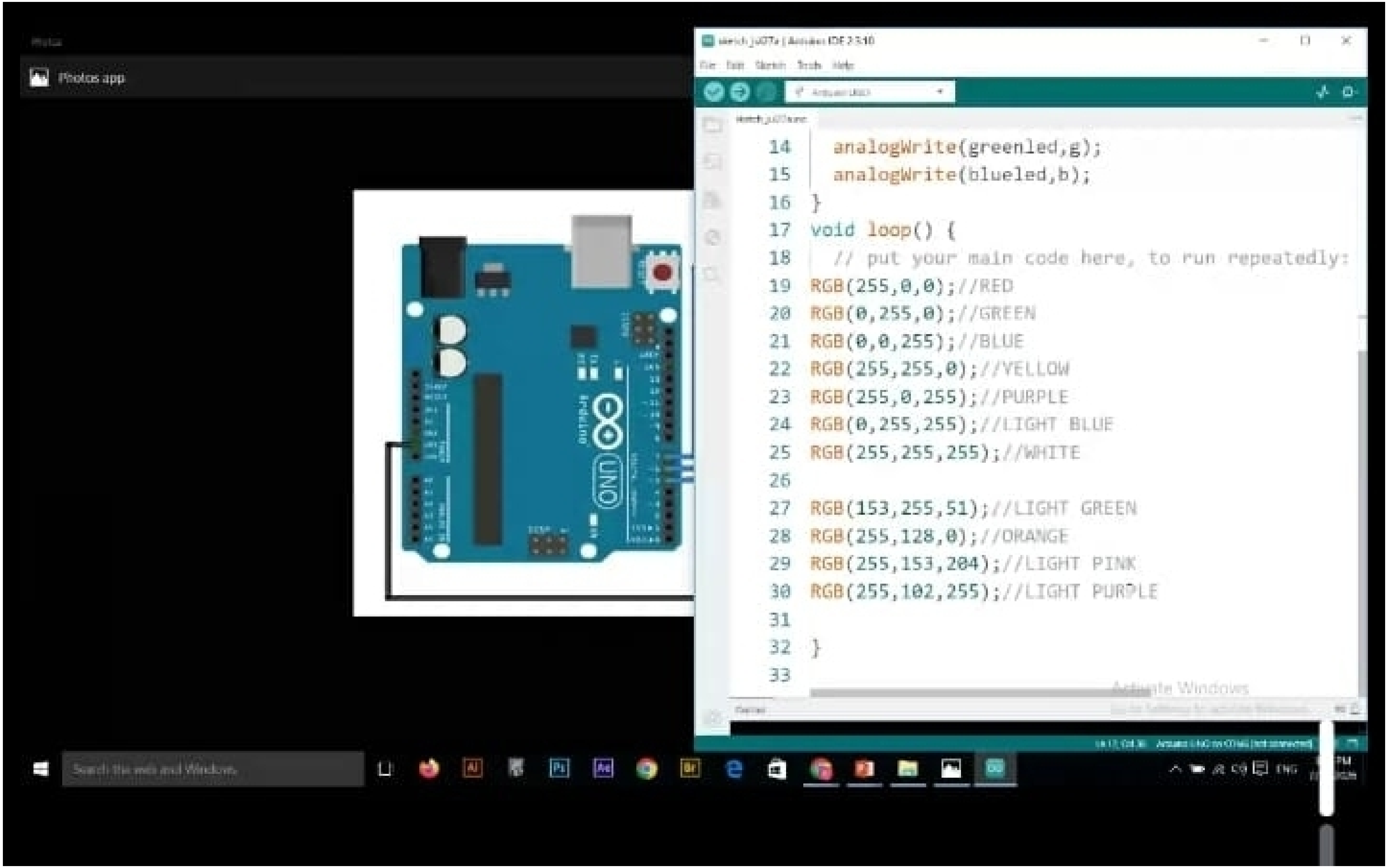

10 - Challenge 1 - Dice Application - PowerPoint (Product Activation Failed)
Animations
Slide Show
Review
View
Developer
Acrobat
Tell me what you want to do...
Text Direction
Align Text
Convert to SmartArt
Arrange
Quick Styles
Shape Fill
Shape Outline
Shape Effects
Find
Replace
Select
Font
Paragraph
Drawing
Editing
Dice Challenge
Required Elements:
Arduino
X 7 Resistor 330 Ω
Red LED
Push Button
Breadboard
dot
Notes
Comments


DIGITAL (PWM~)
UNO
ARDUINO
POWER
ANALOG IN
Notes
Comments


Dice Challenge
Required Elements:
Arduino
X7 Resistor 330 Ω
Red LED
Push Button
Breadboard
dot


Objectives
Made a dice throwing prototype
dot


B 0 V 100 %
80FF00
RGB color codes chart
Hover with cursor on color to get the hex and RGB color codes below:
Hex: # FF66B2
Red: 255
Green: 102
Blue: 178
RGB color space
RGB color space or RGB color system, constructs all the colors from the combination of the Red,
12 void RGB(int r,int g, int b)// r,g&b is a number
13 analogWrite(redled,r);
14 analogWrite(greenled,g);
15 analogWrite(blueled,b);
16 }
17 void loop() {
18 // put your main code here, to run repeatedly:
19 RGB(255,0,0);//RED
20 RGB(0,255,0);//GREEN
21 RGB(0,0,255);//BLUE
22 RGB(255,255,0);//YELLOW
23 RGB(255,0,255);//PURPLE
24 RGB(0,255,255);//LIGHT BLUE
25 RGB(255,255,255);//WHITE
26
27 RGB(153,255,51);//LIGHT GREEN
28 RGB(255,128,0);//ORANGE
29 RGB(255,153,204);
30 RGB(255,102,255);//LIGHT PURPLE
31 RGB(,,);
32 RGB(,,);
33 }
Activate Windows


Photos app
analogWrite(greenled,g);
analogWrite(blueled,b);
}
void loop() {
// put your main code here, to run repeatedly:
RGB(255,0,0);//RED
RGB(0,255,0);//GREEN
RGB(0,0,255);//BLUE
RGB(255,255,0);//YELLOW
RGB(255,0,255);//PURPLE
RGB(0,255,255);//LIGHT BLUE
RGB(255,255,255);//WHITE
RGB(153,255,51);//LIGHT GREEN
RGB(255,128,0);//ORANGE
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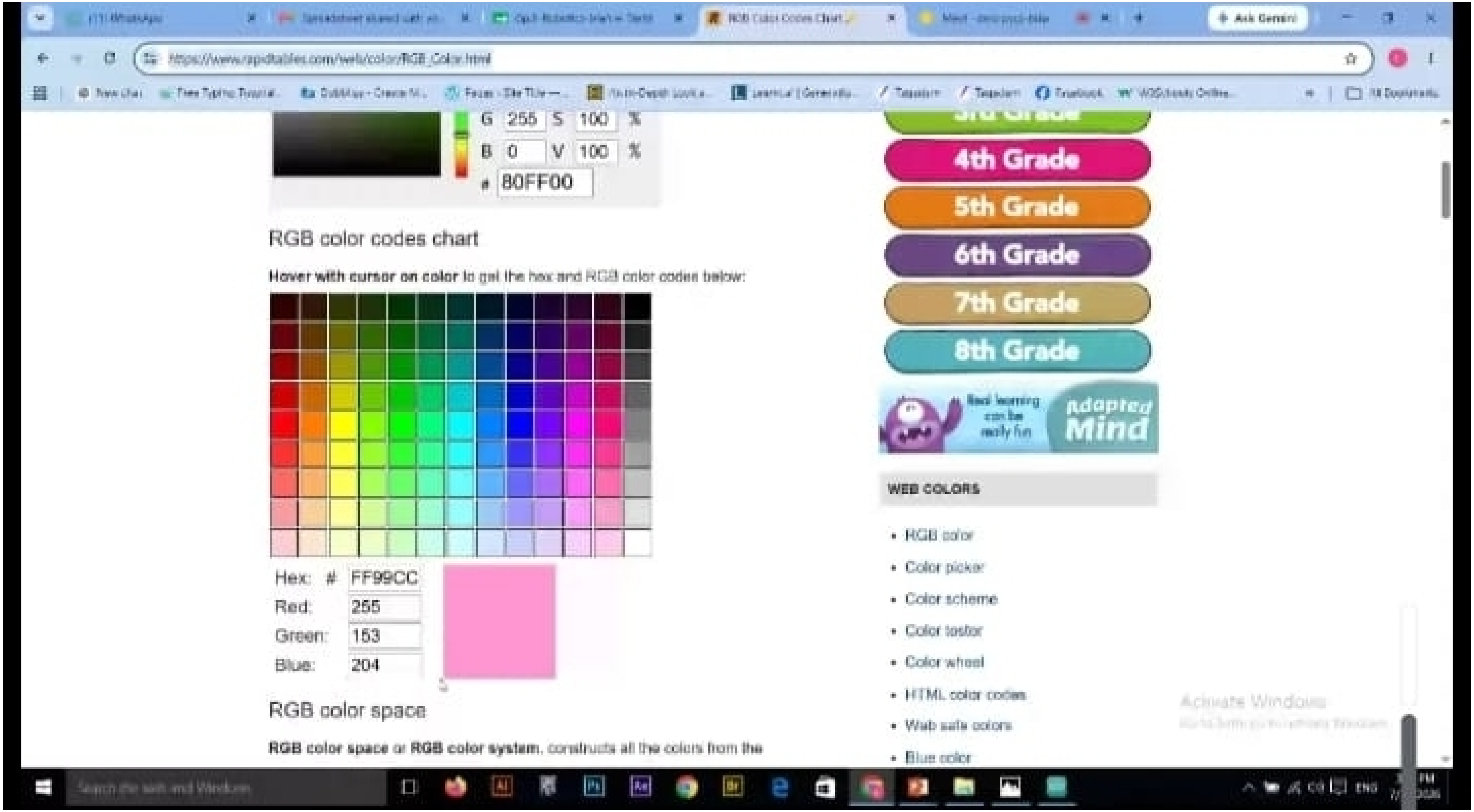
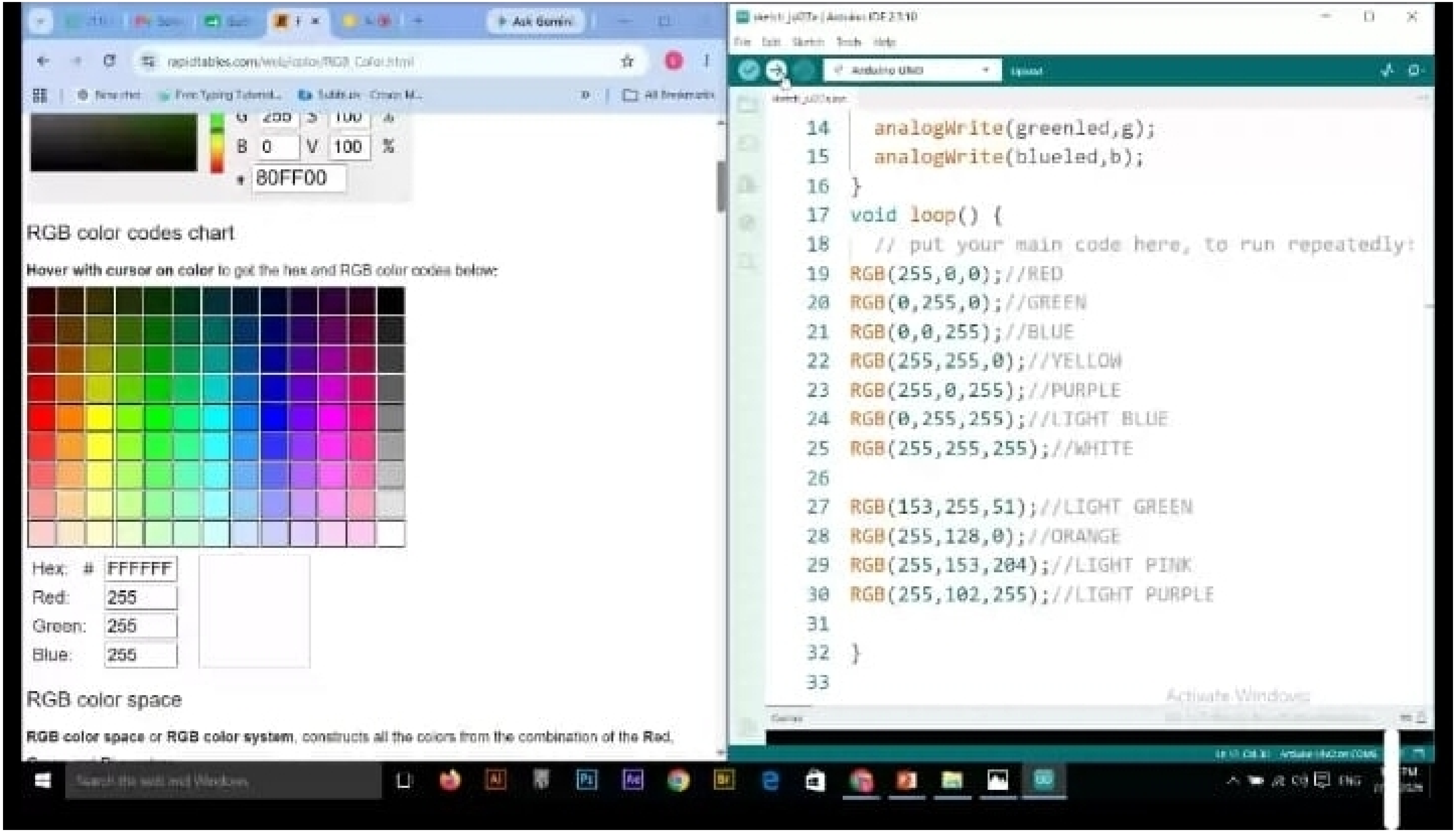
}
UNO


G 255 S 100 %
B 0 V 100 %
80FF00
RGB color codes chart
Hover with cursor on color to get the hex and RGB color codes below:
Hex: # FF99CC
Red: 255
Green: 153
Blue: 204
RGB color space
RGB color space or RGB color system, constructs all the colors from the
4th Grade
5th Grade
6th Grade
7th Grade
8th Grade
Adapted Mind
WEB COLORS
RGB color
Color picker
Color scheme
Color tester
Color wheel
HTML color codes
Web safe colors
Blue color
Activate Windows


Ask Gemini
B 0 V 100 %
80FF00
RGB color codes chart
Hover with cursor on color to get the hex and RGB color codes below:
Hex: # FFFFFF
Red: 255
Green: 255
Blue: 255
RGB color space
RGB color space or RGB color system, constructs all the colors from the combination of the Red,
Arduino UNO
14 analogWrite(greenled,g);
15 analogWrite(blueled,b);
16 }
17 void loop() {
18 // put your main code here, to run repeatedly:
19 RGB(255,0,0);//RED
20 RGB(0,255,0);//GREEN
21 RGB(0,0,255);//BLUE
22 RGB(255,255,0);//YELLOW
23 RGB(255,0,255);//PURPLE
24 RGB(0,255,255);//LIGHT BLUE
25 RGB(255,255,255);//WHITE
26
27 RGB(153,255,51);//LIGHT GREEN
28 RGB(255,128,0);//ORANGE
29 RGB(255,153,204);//LIGHT PINK
30 RGB(255,102,255);//LIGHT PURPLE
31
32 }
33
Activate Windows

```
int redled=9; //ALL THE LEDS SHOULD BE CONNECTED ON PWM PINS(3-5-6-9-10-11)
int greenled=10;
int blueled=11;

void setup() {
  // put your setup code here, to run once:
pinMode(redled,OUTPUT);
pinMode(greenled,OUTPUT);
pinMode(blueled,OUTPUT);
}

void RGB(int r,int g, int b)// r,g&b is a number between 0 and 255{
  analogWrite(redled,r);
  analogWrite(greenled,g);
  analogWrite(blueled,b);
}
void loop() {
  // put your main code here, to run repeatedly:

}
```

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int redled=9; //ALL THE LEDS SHOULD BE CONNECTED ON PWM PINS(3-5-6-9-10-11)
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}

void RGB(int r,int g, int b){
  analogWrite(redled,r); // r is a number between 0 and 255
}
void loop() {
  // put your main code here, to run repeatedly:

}
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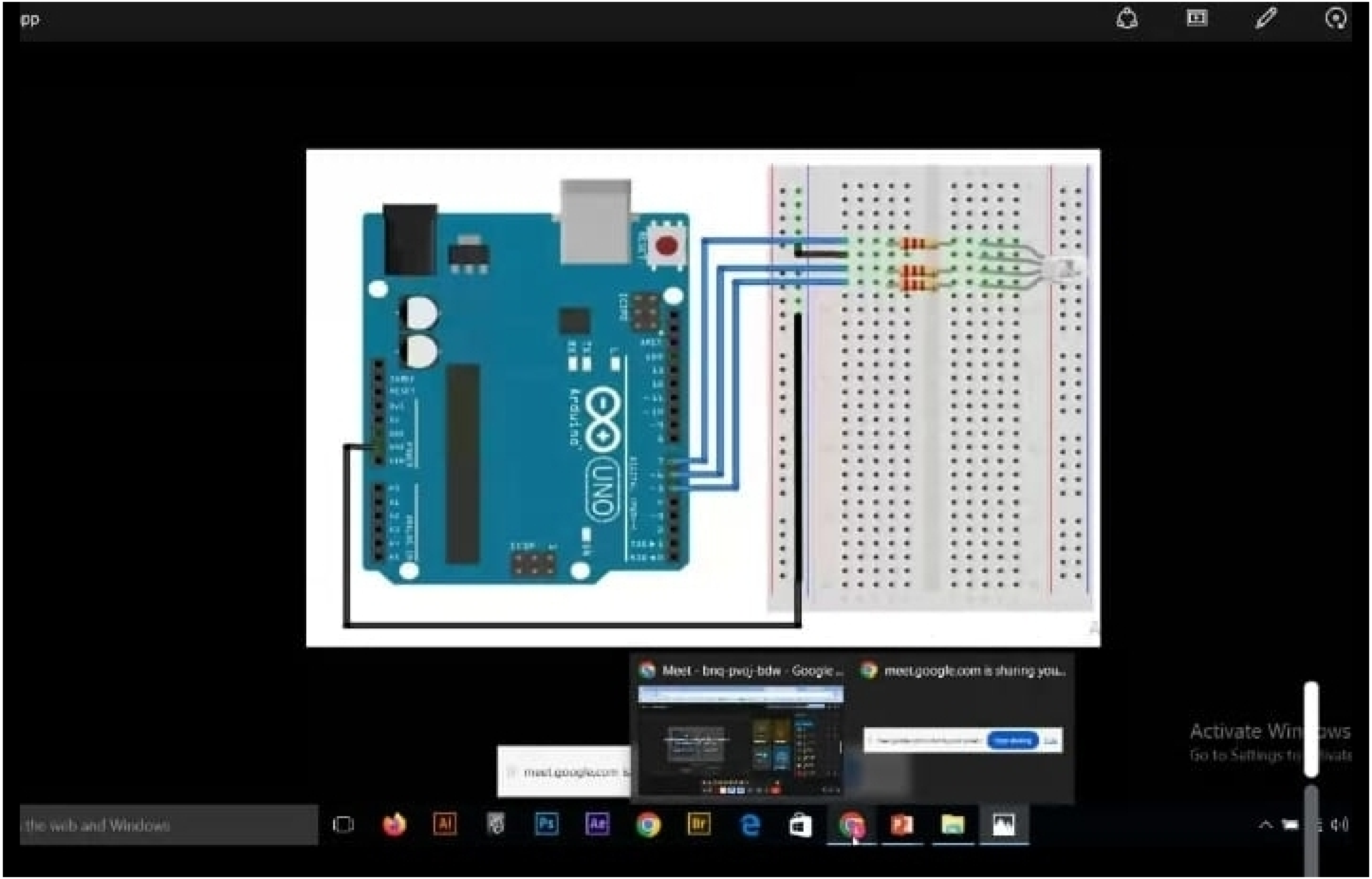
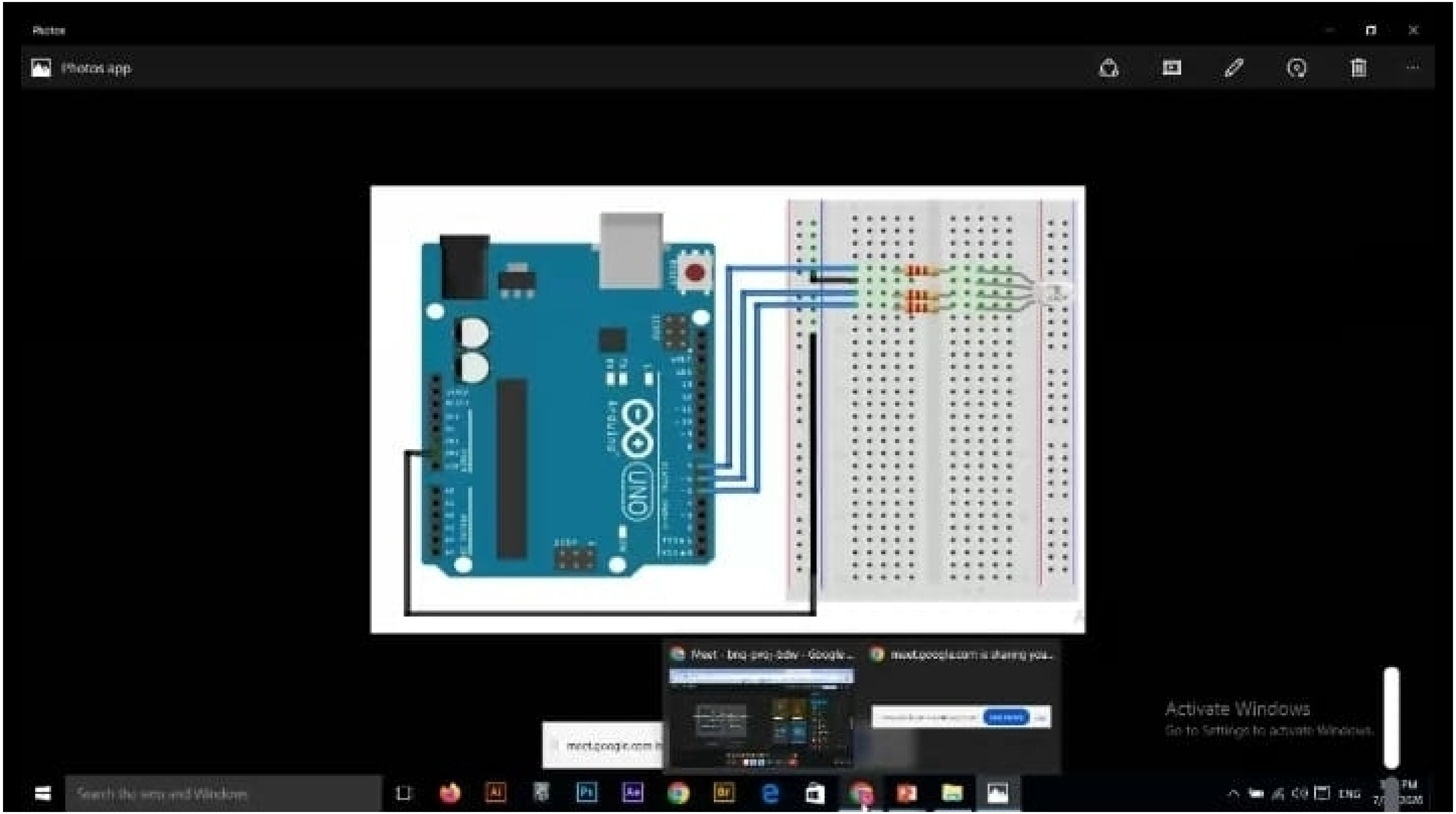
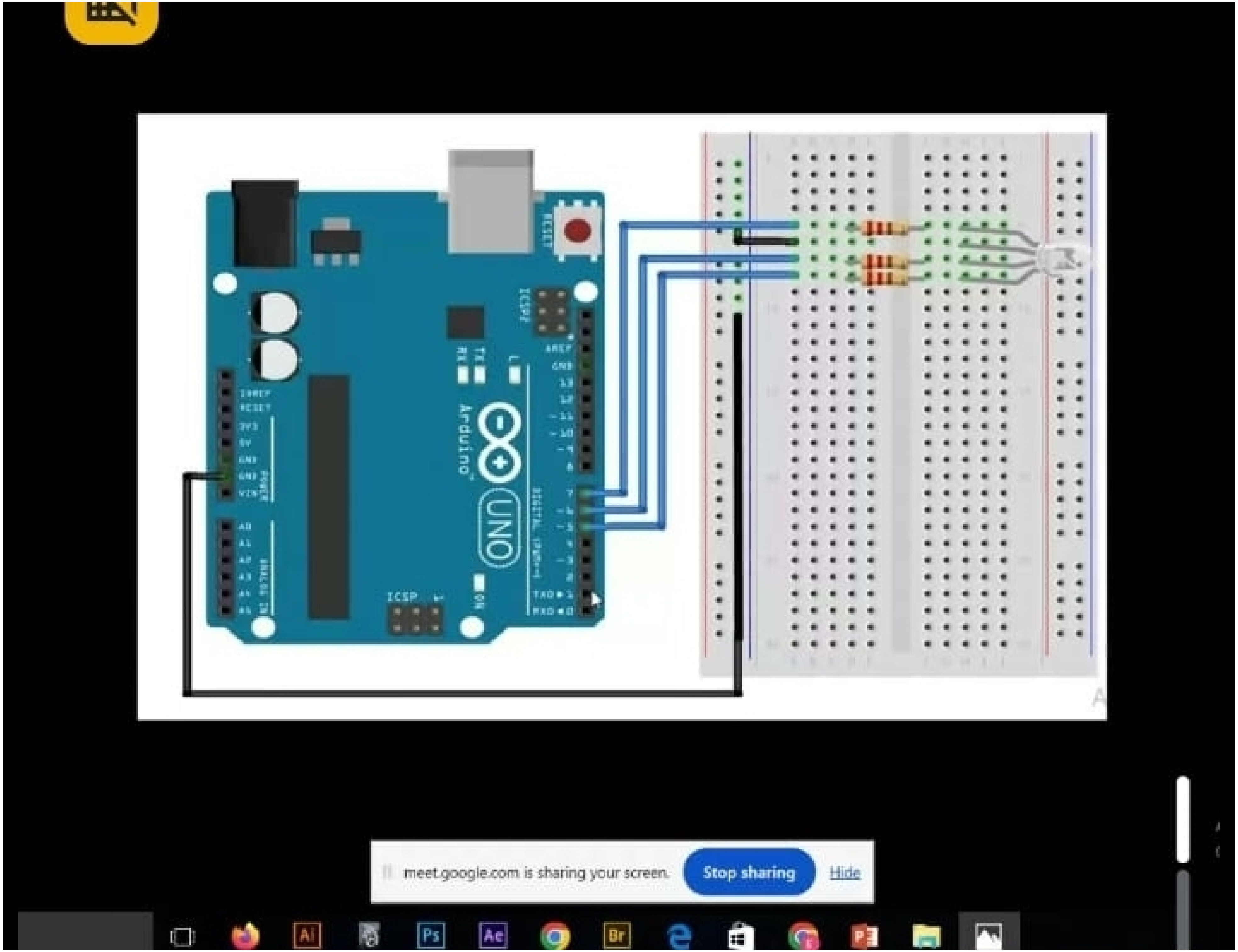
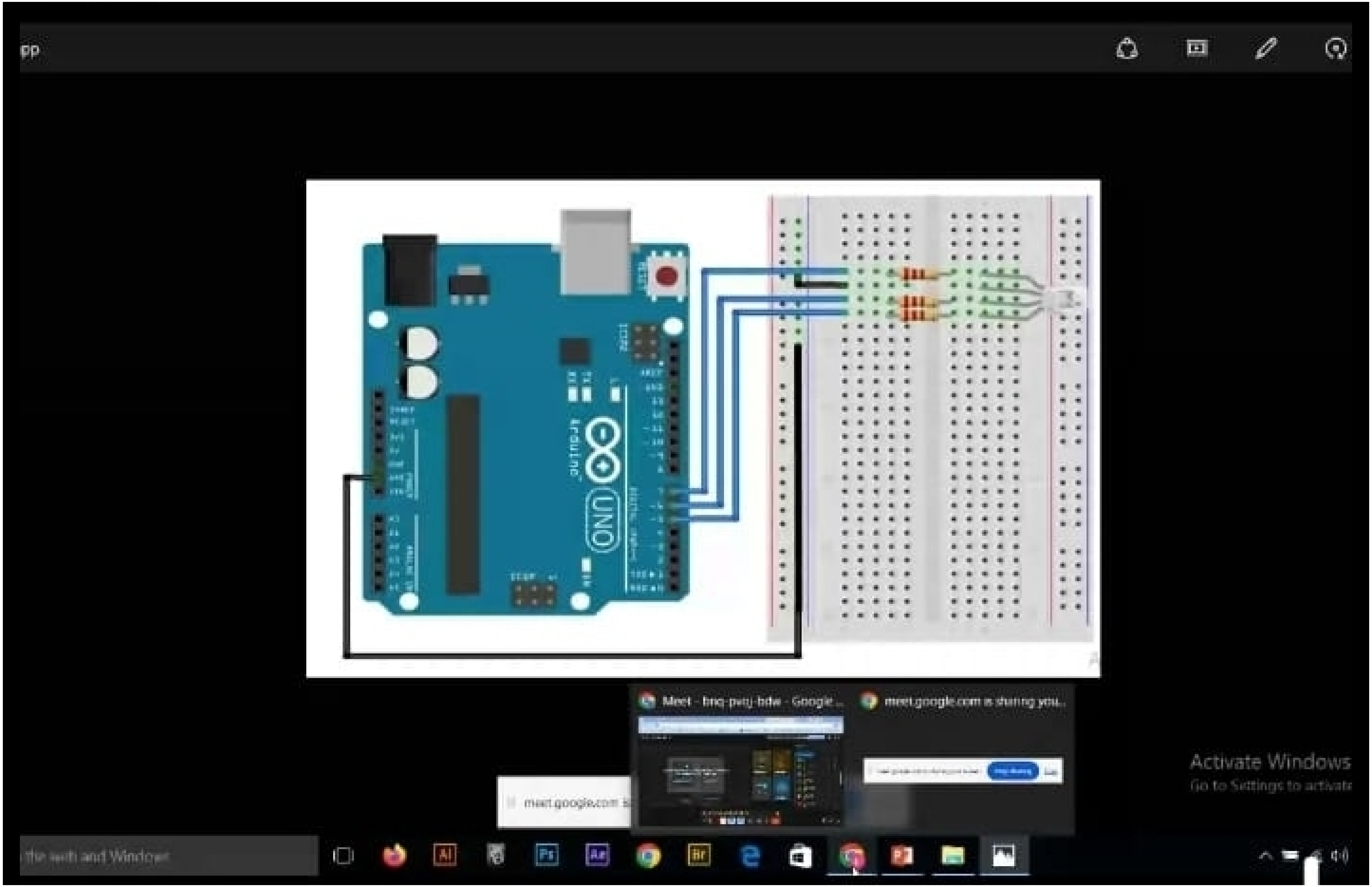


UNO
Arduino
Meet - bnq-pvqj-bdw - Google ...
meet.google.com is sharing you...
Activate Win ows
the web and Windows


Photos
Photos app
UNO
Activate Windows
Go to Settings to activate Windows.
ENG


Arduino
UNO
meet.google.com is sharing your screen.
Stop sharing
Hide


UNO
Meet - bnq-pwq-bdw - Google ...
meet.google.com is sharing you...
Activate Windows
Go to Settings to activate